PHASE 1: Exchange Ingestion, Normalization & Relational Persistence

1. Official Feeds & Sources

Reserve Bank of India
Repo Rate & Inflation

NSE Official Archive
EQUITY_L.csv
2,639 Listed Equities

BSE Scrip Registry
Universal ISIN Matching
72+ Dual Listings

Yahoo Finance Feed
Daily OHLCV Bars
Corporate Action Splits

2. Ingestion & Normalizer

HTTP Stream Ingestion
User-Agent Rotation
Timeout & Retry Guards

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CSV Parser & Sanitizer
UTF-8 Normalization
Type & Date Casting

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ISIN Reconciler
Preserves Price History
Dedupes Dual Listings

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Taxonomy Mapper
Maps 20 NSE Sectors
to Macro Industries

3. PostgreSQL Master (62 Tables)

securities Table
2,639 Listed Equities
ISIN, Ticker & Tiers

instrument_listings
2,640 NSE Listings +
72 Verified BSE Listings

market_prices Table
Adjusted OHLCV Bars
Closing Trade Quotes

fundamental_metrics
ROE, P/E, D/E Ratios
Quarterly Statements

PHASE 2: Fiduciary Candidate Screening Funnel & Optimization Dispatch

4. Fiduciary Candidate Screening Funnel (CandidateUniverseSelector)

Filter 1: Active Status Audit
Screens full 2,639 listed exchange securities.
Excludes 34 delisted, suspended, or BZ stocks.
Result: 2,605 Active Listed Stocks

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Filter 2: Investor Exclusion Mandates
Enforces user-specified negative sectors (e.g. Tobacco, Defense) & banned securities.
Result: Personalized Investment Pool

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Filter 3: Market Price Freshness Audit
Excludes inactive microcaps with stale quotes
(last traded date > 14 calendar days).
Result: Fresh, Liquid Active Securities

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Filter 4: Institutional Liquidity Tiers
Categorizes into High (Nifty 50), Mid (Next 50),
Moderate (Mid/Small 400), and Extended Tiers.
Result: Full Liquidity-Ranked Master

5. Final Recommendation Pool & Optimization Dispatch

Full Active Exchange Recommendation Universe
(Up to 2,605 Liquid Equities)

Passed directly to **13-Node LangGraph Multi-Agent Committee** (Fundamental, Macro, Market, News, Critic) for deep reasoning.

Dispatched to **SciPy SLSQP Solver** for deterministic mathematical weight optimization.

Post-processed with **Whole-Share Floor Rounding**, cash conservation, and statutory Indian friction modeling (STT, Stamp Duty, GST).